

4.3 The impact of tectonic hazards

This chapter will explain:

- ★ the reasons why people live in areas at risk from earthquakes and volcanic eruptions
- ★ the impacts of earthquakes and volcanic eruptions
- ★ how the magnitude of tectonic events is measured.

The reasons why people live in areas at risk from earthquakes and volcanic eruptions

All natural environments provide opportunities and challenges for human activities. Some countries, such as Iceland and the Philippines, were created by volcanic activity. People may choose to live in volcanic areas because they are useful:

- » Volcanic soils are rich, deep and fertile, and allow intensive agriculture to take place.
- » Volcanic areas are important for tourism.
- » Some volcanic areas are seen by people as being symbolic and are part of the national identity, such as Mount Fuji in Japan.

There are several ways of looking at people's vulnerability:

- » One view is that people choose to live in hazardous environments because they understand the environment. In this situation, people choose to live in an area because they feel there are potential advantages to living there and that these outweigh the risks. These advantages include:
 - fertile soils, which increase food production
 - geothermal energy and geothermal spas
 - minerals that can be used in the construction industry
 - volcanoes, which attract tourists
 - employment opportunities in tourism.
- » Another view is that some people live in hazardous environments because they have very little choice over where they live, as they cannot afford to move. Many communities have lived in these areas for generations.
- » Many people may not be aware of the potential risks associated with the hazard. The perception of the risk of the hazard may be low due to:
 - the infrequent occurrence of hazardous events
 - better monitoring, warnings, predictions and building design.

Activity

Suggest contrasting reasons why some people live in volcanically active areas.

The impacts of earthquakes

Hazards can be divided into two types – primary and secondary. Primary hazards are the direct hazards associated with natural events, while secondary hazards are the indirect hazards associated with natural events. [Table 4.2](#) shows the main hazards and impacts associated with earthquake activity.

▼ **Table 4.2** Hazards and impacts associated with earthquakes

Primary hazards	Secondary hazards	Impacts
● Ground shaking ● Surface faulting	● Ground failure and soil liquefaction ● Landslides and rockfalls ● Debris flows and mudflows ● Tsunamis (Figure 4.15)	● Total or partial destruction of building structures ● Interruption of water supplies ● Breakage of sewage disposal systems ● Loss of public utilities such as electricity and gas ● Floods from collapsed dams ● Release of hazardous materials ● Fires ● Spread of chronic illness



▲ **Figure 4.15** The 2004 tsunami caused widespread damage around the countries bordering the Indian Ocean

Activity

Distinguish between primary and secondary hazards, giving an example of each.

The impacts of volcanic eruptions

Likewise, there are several hazards related to volcanic activity ([Table 4.3](#)).

▼ **Table 4.3** The hazards and impacts associated with volcanic activity

Primary hazards	Secondary hazards	Impacts
● Pyroclastic flows ● Volcanic bombs (projectiles) ● Lava flows ● Ash fallout ● Volcanic gases ● Lahars (mudflows) (see Figure 4.9 on page 93) ● Earthquakes	● Atmospheric ash fallout ● Landslides ● Tsunamis ● Acid rainfall	● Destruction of settlements ● Loss of life ● Loss of farmland and forests ● Destruction of infrastructure – roads, airstrips and port facilities ● Disruption of communications

Measuring the magnitude of tectonic events

The magnitude of an earthquake refers to its strength, and it remains unchanged with distance from the epicentre. The intensity describes the shaking caused by an earthquake at a given location, and this decreases with distance from the epicentre. The magnitude of earthquakes is measured using different scales.

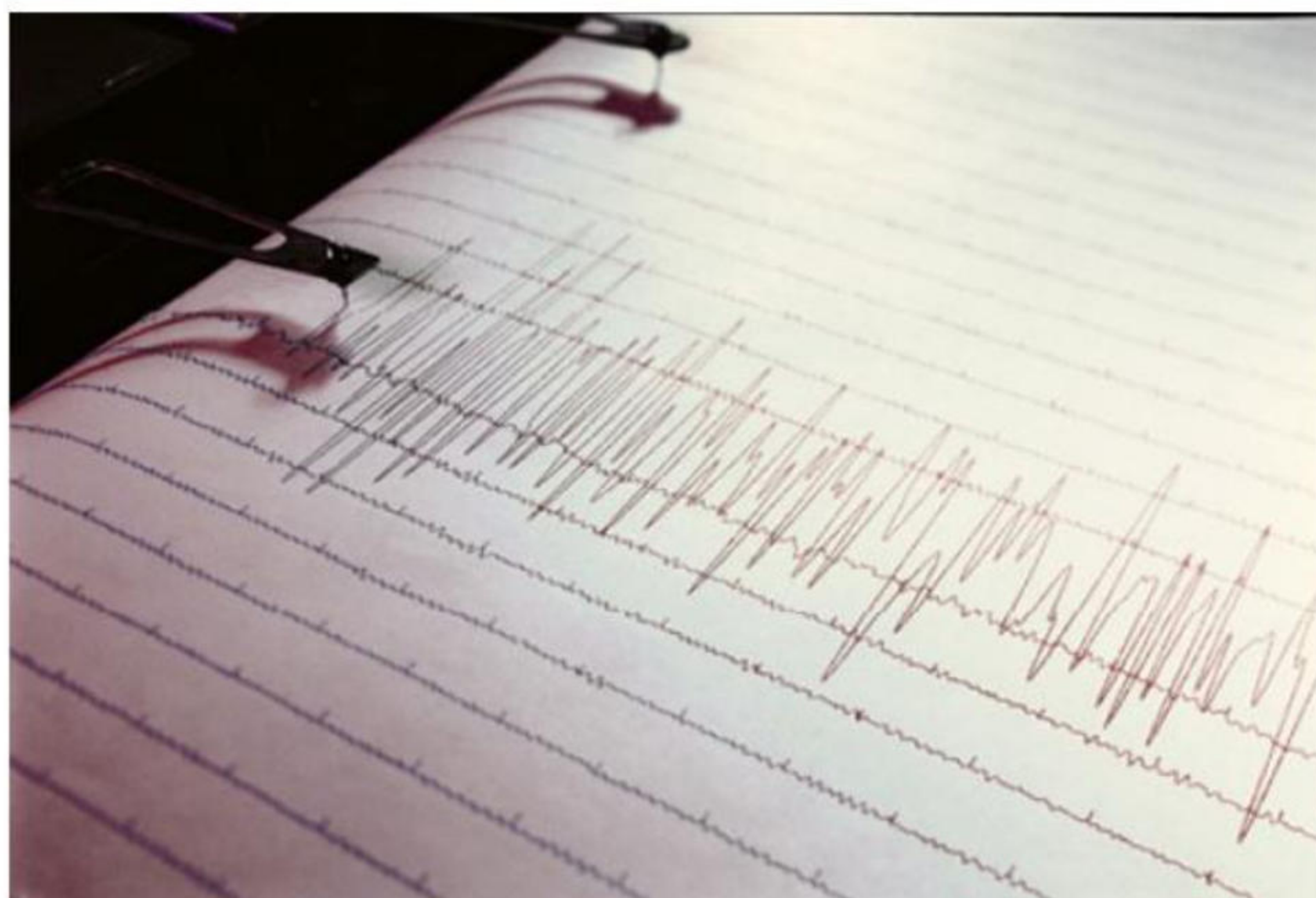
The Richter Scale

In 1935, Charles Richter of the California Institute of Technology developed the **Richter Scale** to measure the magnitude (strength or force) of earthquakes ([Table 4.4](#)). The strength or force of an earthquake is measured on a seismometer and shown on a seismograph ([Figure 4.16](#)). The Richter Scale is logarithmic. This means that an earthquake of 6.0 is ten times greater than one of 5.0, and 100 times greater than one of 4.0.

▼ **Table 4.4** The annual frequency of occurrence of earthquakes of different magnitudes, based on observations since 1900

Descriptor	Magnitude (Richter Scale)	Average number/year	Hazard potential
Great	≥8	1	Total destruction, high loss of life
Major	7.0–7.9	18	Serious building damage, major loss of life
Strong	6.0–6.9	120	Large losses, especially in urban areas

Moderate	5.0–5.9	800	Significant losses in populated areas
Light	4.0–4.9	6200	Usually felt, some structural damage
Minor	3.0–3.9	49,000	Typically felt, but usually little damage
Very minor	<3.0	9000/day	Not felt, but recorded



▲ **Figure 4.16** A seismograph reading from Soufrière Hills, Montserrat

The Mercalli Scale

In contrast, the **Mercalli Scale** ([Table 4.5](#)) relates ground movement to things that you would notice happening around you. It is a subjective scale and lacks accuracy/reliability about the magnitude of the earthquake event. However, its main advantage is that it allows ordinary eyewitnesses to provide information on the strength of an earthquake. For example, a weak earthquake (II on the scale) might be felt by only a few people at rest in the upper floors of a building, whereas a violent earthquake (IX on the scale) is likely to cause considerable damage to buildings, with partial collapse.

▼ **Table 4.5** The Mercalli Scale

Strength	Observations
I	● Rarely felt
II	● Felt by people who are not moving, especially on the upper floors of buildings ● Hanging objects may swing
III	● The effects are noticeable indoors, especially upstairs ● The vibration is like that experienced when a truck passes
IV	● Many people feel it indoors; a few outside ● Some are awakened at night ● Crockery and doors are disturbed, and standing cars rock
V	● Felt by nearly everyone – most people are awakened ● Some windows are broken, plaster becomes cracked and unstable objects topple ● Trees may sway and pendulum clocks stop
VI	● Felt by everyone – many are frightened ● Some heavy furniture moves, plaster falls ● Structural damage is usually quite slight
VII	● Everyone runs outdoors ● Noticed by people driving cars ● Poorly designed buildings are damaged
VIII	● Damage to ordinary buildings – many collapse

	● Well-designed buildings survive but suffer slight damage ● Heavy furniture is overturned and chimneys fall
IX	● Damage occurs even to buildings that have been well designed ● Many buildings are moved from their foundations ● Ground cracks and pipes break
X	● Most masonry structures are destroyed; wooden ones may survive ● Railway tracks bend and water slops over banks ● Landslides and sand movements occur
XI	● No masonry structure remains standing; bridges are destroyed ● Large cracks occur in the ground
XII	● Total damage ● Waves are seen on the surface of the ground and objects are thrown into the air

The Moment Magnitude Scale

Scientists are increasingly using the **Moment Magnitude Scale** (M), which measures the amount of energy released in an earthquake. For every increase of 1 on the M scale, the amount of energy released increases by more than 32. Every increase of 0.2 represents a doubling of the energy released. Although this may be a more reliable and scientific measurement, a subjective scale may be enough for those who are affected by an earthquake, and the media still generally refer to the Richter Scale.

The Volcanic Explosivity Index

The strength of a volcano is measured by the **Volcanic Explosivity Index** (VEI) ([Figure 4.17](#)). This is based on the amount of material ejected in the explosion, the height of the cloud it caused, and the amount of damage that results. Any explosion above level 5 is considered to be very large and violent.

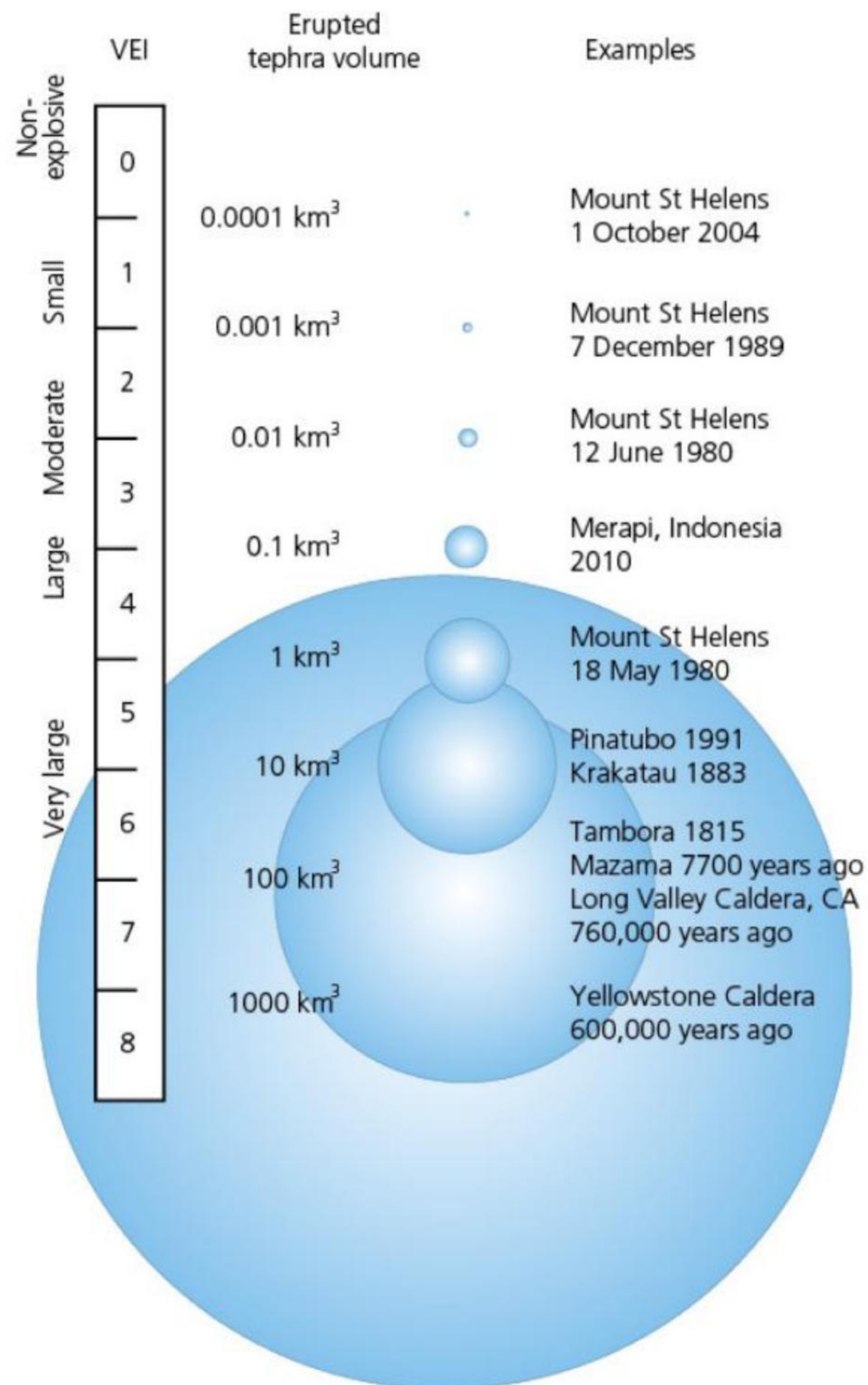
A **supervolcano** is a volcano of VEI 8. The scale is logarithmic, so VEI 8 is ten times more powerful than VEI 7, 100 times more powerful than VEI 6, and 1000 times more powerful than VEI 5 (Mount Pinatubo, 1991). The last VEI 8 was 74,000 years ago (Mount Toba, Indonesia).

Supervolcanoes tend to be much larger than 'normal' volcanoes – the Yellowstone magma chamber, for example, is over 50 km wide. The likely impacts of a VEI 8 eruption include:

- » almost complete loss of life within about 1000 km of the eruption
- » destruction of all crops and livestock, leading to a global famine
- » economic and social devastation.

Activities

- 1 Distinguish between earthquake magnitude and intensity.
 - 2 The Richter Scale is logarithmic. How much stronger is an earthquake of 7.0 on the Richter Scale compared with one of 5.0?
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▲ **Figure 4.17** The Volcanic Explosivity Index